

Biotechnology And Its Applications

INTRODUCTION

Biotechnology : deals with industrial scale production of biopharmaceuticals and biologicals using genetically modified microbes, fungi, plants & animals

-Application includes therapeutics, diagnostics, genetically modified crops for agriculture, processed food, bioremediation, waste treatment, & energy production

-Human beings have used biotechnology to improve the quality of human life especially in the field of food production & health

Three Critical Research areas of Biotechnology :

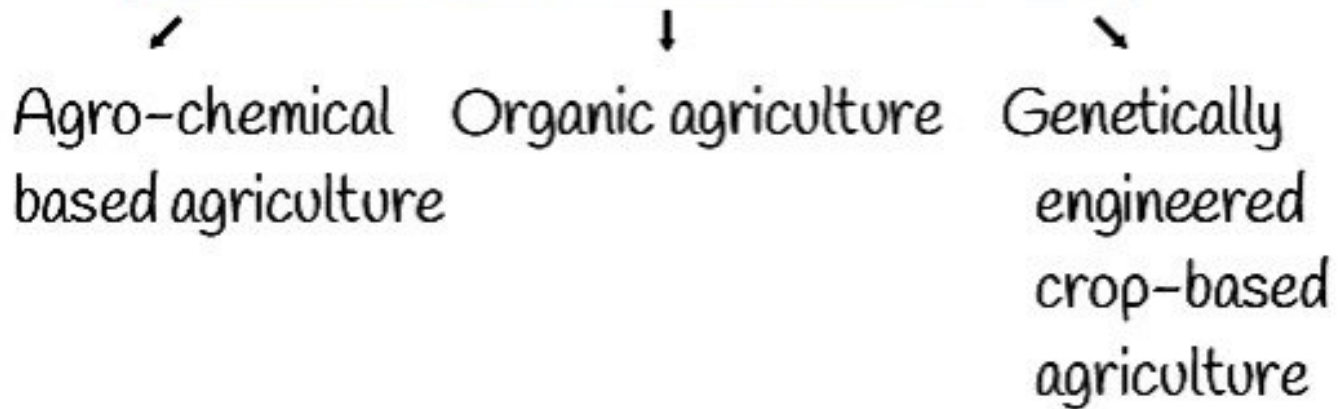
- (a) Providing best catalyst in the form of improved organism usually a microbe or pure enzyme
- (b) Creating optimal conditions through engineering for a



catalyst to act

(c) Downstream processing technologies to purify protein / organic compounds

Biotechnology Application In Agriculture



GREEN REVOLUTION

- Tripled food supply
- Increased yields partly due to use of improved crop varieties
- Mainly due to better management practices

↓
And use of **Agrochemicals**

↓
Expensive

~Plants, bacteria, fungi and animals whose genes have



been altered by manipulation are called Genetically modified organisms (GMO)

GENETIC MODIFICATION HAS -

- Made crops more tolerant to abiotic stresses
- Reduced reliance on chemical pesticides
- Helped to reduce post-harvest losses
- Increased efficiency of mineral usage by enhanced nutritional value of food

Eg : Vitamin 'A' enriched rice

- Enhanced nutritional value of food

Eg : Golden rice

Genetic Modification

- Tailor-made plant to supply Alternative resources to industries, in the form of Starches, fuels & Pharmaceuticals

Pest Resistant Crops

✓ Insect resistant Plant

✓ Nemotode resistance Plant



INSECT - RESISTANT PLANTS

During a particular stage
Strains of Bacillus thuringiensis → **Proteins** ↗



Kills insects such as

- Lepidopterans (Tobacco budworm, armyworm)
- Coleopterans (Beetles)
- Dipterans (Flies, Mosquitoes)

Bt Toxin → Inactive → Ingested by → Converted
protein protoxins insects into active
Form

(Due to alkaline PH of gut)



Death of Insects



Swelling & Lysis ← Create Pores ← Epithelial cells

BT - Crystal (Cry protein) ↗

Toxic to larvae of certain insects



BT – Crystals (Cry Protein)

↓ gene

"Cry gene"

↓

Cry IAc & Cry IIAb control cotton bollworms

Cry IAb controls Cron borer

PEST RESISTANT PLANTS

- Nematode *Meloidogyne incognita* infects the roots of tobacco plants
- Process of RNA interference was adopted to prevent the infection of tobacco plants

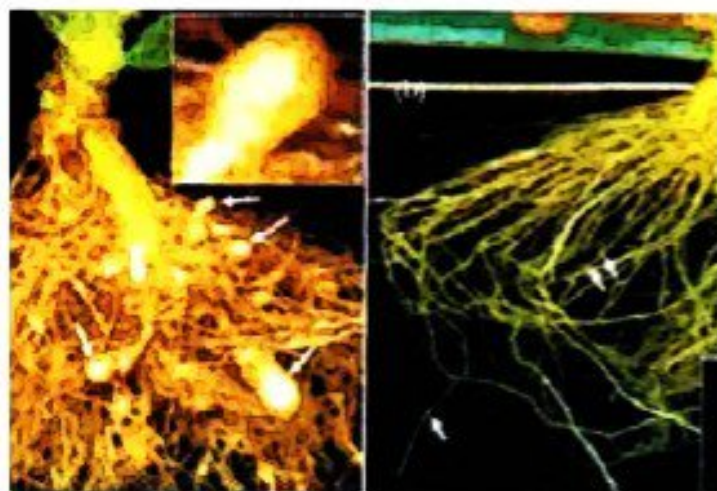
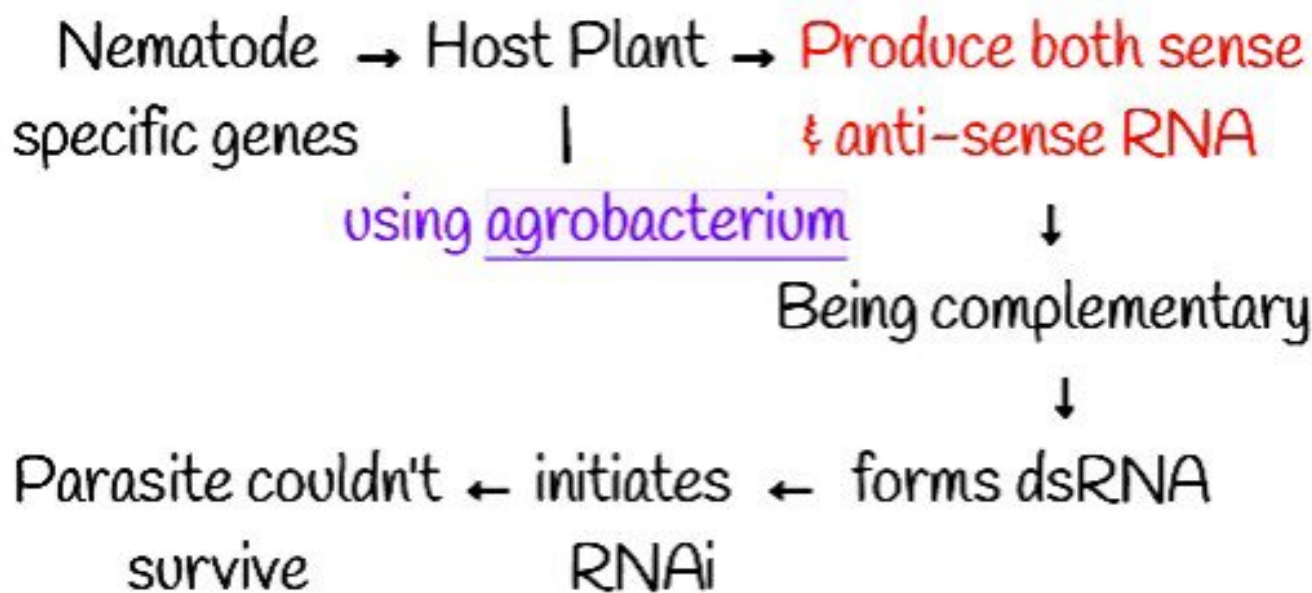
RNA interference (RNAi) involves :

- Silencing of a specific mRNA due to a complementary dsRNA (double stranded RNA) molecule
- Takes place in eukaryotes as a method of cellular defence

Source of Complementary RNA



From an infection by viruses having RNA genome /
mobile genetic elements



(a) self-generated dsRNA triggers protection against nematodes of a typical control plants; (b) transgenic plant roots free of nematode but protected through novel mechanism



Figure-12.2

Biotechnological Applications in Medicine

- Recombinant DNA technological processes have made immense impact in the area of healthcare by enabling mass production of genetically engineered medicines such as insulin, by creating methods like gene therapy, recombinant DNA technology, polymerase Chain Reaction (PCR) and Enzyme Linked Immuno-sorbent Assay (ELISA)
- About 30 recombinant therapeutics have been approved for human-use over the world

Genetically Engineered Insulin :

- (1) Genetically engineered Insulin — Proteinaceous Hormone
Having 51 amino acids



A-chain

B-chain



21

30

S-S Disulphide Bond

Banting and Best

-First to isolate insulin from Dog pancreas

Sir Edward Sharpy-Shafer

-Diabetes due to failure of some islands of pancreas

First genetically engineered insulin

By recombinant DNA technique → Using E. coli



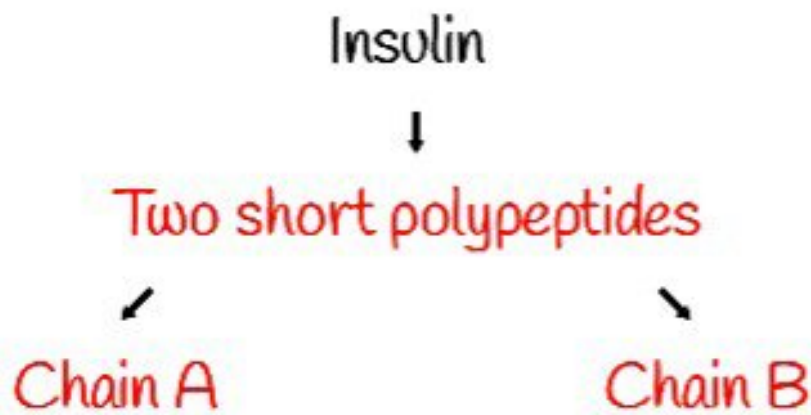
Eli Lilly ← By American firms



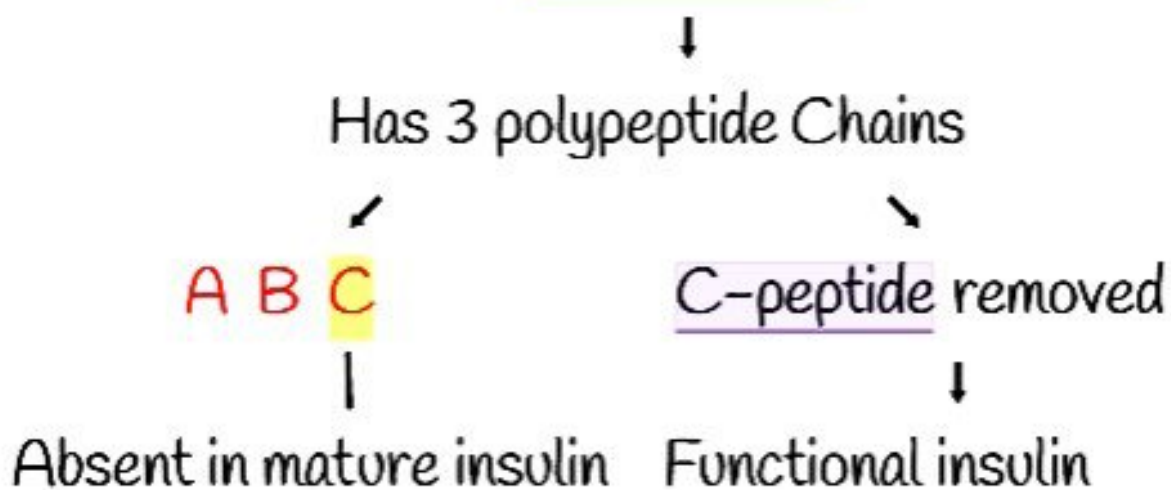
Called it Humulin

Insulin earlier extracted from pancreas of slaughtered cattle & Pigs - Caused allergy & Other reactions





In mammals, including humans, insulin is synthesised as pro hormone



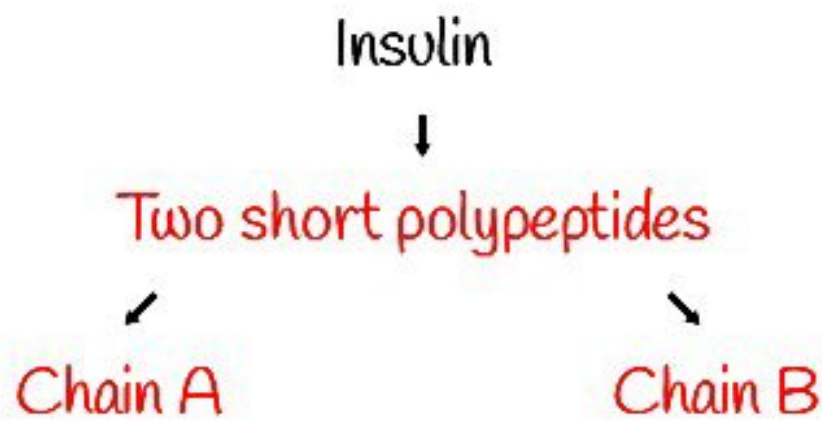
In 1983, Eli Lilly

Prepared two DNA sequence corresponding to A & B chains

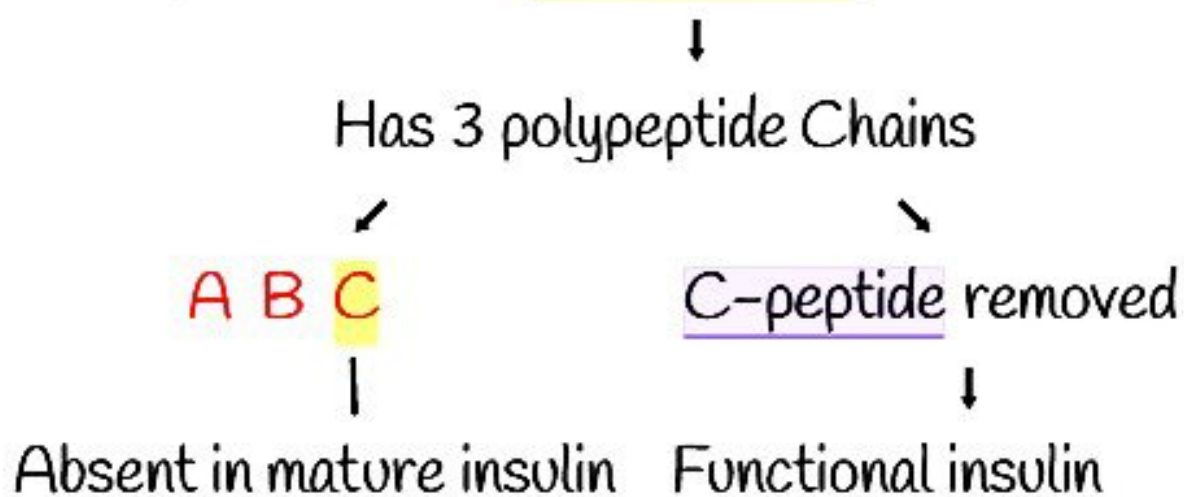
↓

Introduced them in plasmids of E.coli - Produce





In mammals, including humans, insulin is synthesised as pro hormone



In 1983, Eli Lilly

Prepared two DNA sequence corresponding to A & B chains

↓

Introduced them in plasmids of E.coli - Produce



Insulin



Extracted ← Chains A & B produced separately

Combined by disulphide bond

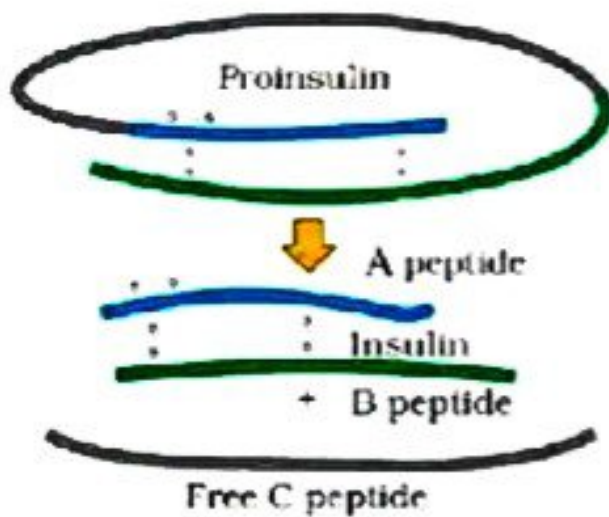


Figure 12.3 Maturation of pro-insulin into insulin (simplified)

Figure-12.3

Take function of non-functional gene



Involves delivery of a normal gene into



Individual or embryo



Correction



Treat disease ← Gene Therapy → gene-person's
such as SCID, cells



Haemophilia Collection of methods
that allow correction of
a gene defect that has been
Diagnosed in a child / embryo

First Clinical Gene Therapy

1990 to a 4-year old girl



ADA deficiency (adenosine deaminase)



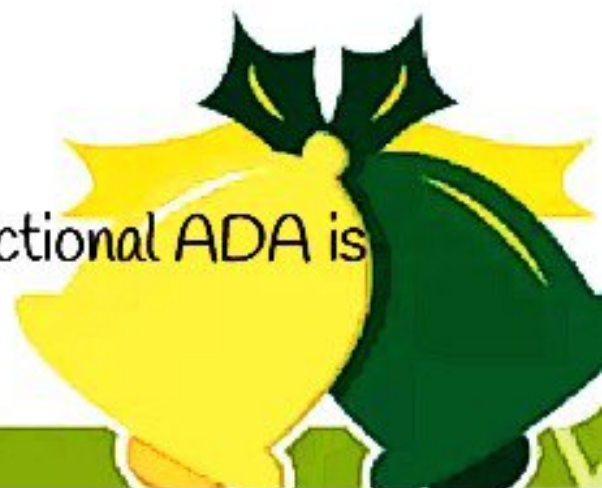
Due to deletion of gene
for adenosine deaminase



Crucial for immune
system to function

Ways to Cure

- Bone marrow transplantation
- Enzyme replacement therapy – functional ADA is



given to the patient by injection

- Lymphocytes from blood of patients grown in culture outside the body

Functional ADA cDNA → Introduced → subsequently
(Using retroviral vector) into lymphocytes returned
If inserted at an early stage to patient
Permanent cure (requires periodic Infusion)

MOLECULAR DIAGNOSIS

For effective treatment - early diagnosis & understanding it's pathophysiology is important

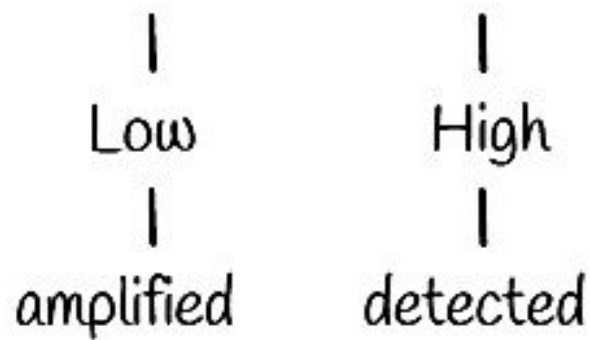
Conventional Methods

- Early detection not possible
- (Sperm & urine analysis)

Modern Methods

- (rDNA technology, PCR, ELISA)
- Early detection possible
- Presence of pathogen





PCR (Polymerase Chain Reaction)

- Detects HIV in AIDS patients
- Used to detect mutations in Cancer patients
- Powerful techniques to identify genetic disorders

Antigen-antibody interaction



Principle



ELISA (Enzyme linked immuno-sorbent assay)



Infection detected by presence of Antigens



(Glycoproteins, proteins)

Animals genes manipulated to possess
& express an extra gene



95% are mice ← Transgenic Animals



Eg : Transgenic rats , rabbits , pigs
Sheep , cows & fish

(1) Normal Physiological and Development

Transgenic animals are desired to study how genes are regulated & affect the normal functions of body & development

Eg : Study of complex factors involved in growth such as Insulin-like growth factors

(2) Study of Diseases

Transgenic animals are designed to increase our understanding of how genes contribute to the development of disease

- Specially serve as model for human diseases
- So , investigation for new disease is possible
- Transgenic models exist for many human diseases such



as Cancer , Cystic fibrosis , rheumatoid arthritis & Alzheimer's

(3) Biological Products

To produce biological products such as human protein (α -1-antitrypsin) used to treat emphysema, proteins to treat phenylketonuria & cystic fibrosis, human alpha-lactalbumin enriched milk

Eg : Example - (1997) The first transgenic cow ROSIE - produced human protein-enriched milk
2.4 grams per litre

|

- Contains human α -lactalbumin
- Nutritionally more balanced product for human babies than natural cow-milk

(4) Vaccine Safety

- For testing the safety of vaccines before they are used on humans
- Example - Transgenic mice are being used to test the safety of the polio vaccine



-If successful, can replace use of monkeys

(5) Chemical Safety Testing

- Known as toxicity / safety testing
- Procedure same as toxicity of drugs
- Transgenic animals carry genes which make them more sensitive to toxic substances than non-transgenic animals
- Allows us to obtain results in lesser time

ETHICAL ISSUES

GEAC (genetic engineering approval committee)

- (1) Indian government organisation
- (2) Makes decisions regarding the validity of GM research & the safety of introducing GM organisms

BIO-PATIENT

Patient right granted by a government to an
Inventor

Is Granted for

- An Invention
- An improvement to an earlier invention
- A new concept or design



RICE

- Important food grain
- Presence goes back thousands of years in Asia's agricultural history
- Estimated 2,00,000 varieties
- Diversity in India – richest in world
- **Basmati Rice**

Distinct for its unique aroma

Reference in ancient texts, folklore & poetry

- In 1997, an American company got patent rights

'New' variety of Indian basmati crossed with semi-dwarf variety

- Several attempts have also been made to patent uses, products & processes based on Indian traditional herbal medicines Example – turmeric, neem

BIOPIRACY

- Biopiracy is the term used to refer to the use of



bio-resources by multinational companies and other organizations without proper authorization from the countries and people concerned without compensatory payment

-Some nations are developing laws to prevent such unauthorized exploitation of their bio-resources and traditional knowledge

Indian Parliament



Cleared the 2nd amendment of indian patent Bill



Takes issues into consideration, including patent items, emergency funds & research development Initiatives

Industrialized Nations



Rich financially but poor in diversity & traditional Knowledge

